

Pranav Viswanathan

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EDUCATION

Northeastern University

Master of Science in Computer Science

Boston, MA

Sept 2024 – May 2026

Vellore Institute of Technology

Bachelor's in Technology in Computer Science and Engineering

Chennai, IN

Sept 2020 – May 2024

EXPERIENCE

SDE Intern

May 2023 – Jul 2023

NergyLive

Remote

- Built real-time HVAC anomaly detection using isolation forests and statistical process control, processing 10K+ sensor readings/minute with sub-100ms latency
- Built interactive Plotly dashboards with drill-down capabilities for 20+ system parameters, enabling engineers to identify root causes 3x faster, reducing troubleshooting time from 45 to 15 minutes
- Deployed anomaly detection system into production in collaboration with cross-functional teams, reducing downtime incidents and cutting maintenance costs by 15%

PROJECTS

Lucera | Python, FFmpeg, OpenCV, Docker, Shell Scripting

Dec 2025 – Jan 2026

- Architected modular video processing pipeline with 5 independent stages (analysis, enhancement, packaging, verification) handling 4K video processing with automated error recovery and logging
- Implemented adaptive bitrate HLS streaming with FFmpeg, generating multiple quality variants (360p-4K) and reducing bandwidth usage by 40% while maintaining playback quality across network conditions
- Built quality verification system using VMAF, PSNR, and SSIM metrics to ensure 95+ perceptual quality scores, with OpenCV-based scene detection analyzing blur, noise, and motion complexity
- Containerized entire pipeline with Docker and automated deployment through shell scripts, enabling reproducible builds

Rebalance-AI | Python, Apache Airflow, Docker, PostgreSQL, Redis, GitHub Actions

Sept 2025 – Dec 2025

- Built automated ETL pipeline processing 3.16M records with sub-10 minute runtime using Apache Airflow DAGs, geospatial indexing, and optimized PostgreSQL queries across 400+ bike stations
- Reduced deployment time from 2 hours to 15 minutes by implementing Docker containerization with multi-stage builds and GitHub Actions CI/CD for automated testing and rollback mechanisms
- Developed real-time monitoring dashboard tracking 15+ metrics (pipeline latency, data quality, resource usage) using MLflow telemetry and Prometheus with custom alerting
- Improved API throughput by 40% to 1000+ requests/minute with sub-200ms P95 latency through Redis caching, load balancing, and asynchronous processing
- Presented technical architecture and scalability approach to engineering teams at Google Cambridge

HawkEye | Python, OpenCV, PyTorch, Docker, Shell Scripting

Sept 2025 – Nov 2025

- Built real-time multi-sensor processing pipeline handling RGB, thermal, and LiDAR streams with sub-100ms latency achieving 14-16 FPS throughput
- Implemented Kalman Filter object tracking with IoU association, maintaining track continuity across 400+ frames with automated state estimation
- Developed sensor simulation framework with configurable degradation effects (fog, noise, jamming) for testing system robustness under edge cases
- Created automated video processing pipeline with JSON logging for object detections, timestamps, and trajectory tracking across video sequences

TECHNICAL SKILLS

Languages: Python, Java, Go, C/C++, JavaScript, SQL, HTML/CSS, Shell Scripting

Frameworks & Libraries: Flask, FastAPI, React, Apache Airflow, PyTorch, OpenCV, pandas, NumPy, scikit-learn

Developer Tools: Git, Docker, GitHub Actions, terraform, Redis, MLflow, Prometheus, FFmpeg, VS Code, PyCharm, IntelliJ, Locust

Databases & Cloud: PostgreSQL, Redis, AWS (S3, EC2, Lambda), Google Cloud Platform, Docker Containerization

Concepts: CI/CD Pipelines, RESTful APIs, Microservices, Data Structures & Algorithms, System Design, MLOps